Report Biomimicry Challenge

ZHAW Institute for Sustainable Development | Blockweek Bionics 2026

Challenge 1 — Urban Heat Islands

Cooling the City

“Picture”

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Students	**Contribution to the report / (sections)**
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[TODO: Name 3]	**[TODO: e.g. Sections 1, 5, figures and editing]**
[TODO: Name 4]	**[TODO: e.g. Section 3.3, measurements, appendix]**

Comments	**[TODO: optional remarks for the instructors]**

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Contents

1 Introduction

1.1 Background and relevance

Impervious urban surfaces absorb most of the incoming short-wave radiation and store it, sealing removes the evaporative cooling that vegetation would provide, and the narrow geometry of a street canyon keeps the emitted long-wave radiation from escaping to the sky (Oke 1982). Zurich measures the result as an air-temperature difference of 1 K to 2 K by day and of up to 7 K at night against the surrounding area, with sun-exposed sealed surfaces above 60 °C in summer (Stadt Zürich 2021). The stored energy is released into the street canyon, and that is where the load is felt: a surface at 60 °C radiates at eye level onto anyone walking, waiting or sitting in the street, while the same person indoors is shielded from it and can ventilate at night. Public outdoor space is the part of the city that cannot be conditioned, and it is where the city's health concern applies — cerebrovascular, cardiovascular and respiratory illness during heat periods, with older people and small children most at risk (Stadt Zürich 2021).

Air conditioning is no answer to this: it rejects the heat into the street as waste heat and raises electricity demand at the peak hour, which is why the city advises against it (Stadt Zürich 2020a, pp. 132–133). What Zurich pursues instead — the *Fachplanung Hitzeminderung* with its thematic maps and implementation agenda (Stadt Zürich 2020a; Stadt Zürich 2020b; Stadt Zürich 2020c) — acts on the outdoor space at pedestrian level. That space is bounded by pavement and by the façades of a building stock that already stands, so those surfaces are the intervention point a retrofit can reach: the target is the street in front of the façade, not the room behind it. Section 2.3 sets out what the city specifies for them today, and with it the benchmark this project has to beat.

1.2 Aim of the report and intended contribution

This report documents a biomimicry design process for reducing heat load on the envelope of existing urban buildings, and with it on the street space those envelopes enclose. It follows the Biomimicry Design Spiral (Baumeister et al. 2014) through scoping, biological discovery and abstraction, engineering concept development, validation and evaluation, in the challenge-to-biology direction described by Helms et al. (2009). The intended contribution is a technically specified, quantitatively checked envelope concept, together with a transparent account of which biological mechanisms were transferred, which were rejected, and on what evidence.

1.3 Structure of the report

This report is organised as the reasoning chain it documents, and it is written so that the chain can be followed backwards as well as forwards: challenge → function → biology → abstraction → concept → engineering check → evaluation → iteration.

Section 2 cuts the given challenge down to something answerable. It fixes a reference city and a reference site, states the design question and the four candidates rejected to reach it, draws the system boundary, lists the functions as solution-neutral verbs and turns them into measurable criteria. Section 3 carries out the biological search those functions define, abstracts the mechanisms found into transferable principles, develops the concepts, and tests them. Section 4 assesses the result against Life's Principles rather than listing them, and Section 5 asks what the biomimicry route contributed that a conventional one would not have, and where the evidence runs out.

Because the method is judged as much as the result, **every substantive decision in this report is stated together with the alternative it displaced and the reason.** Those decisions are argued where they arise, and collected in one place in Appendix B, which can be read as a map of how the project reached its current shape.

2 Challenge & Scoping

2.1 Problem definition and design question

Given challenge (Challenge 1). *Cities are facing increasing heat stress due to climate change. Develop bioinspired strategies to reduce urban heat through building materials, façade design, or urban planning.*

The challenge as given spans three scales — material, building component and urban planning. Working on all three would make the project larger rather than more focused, which the systems view is explicitly meant to prevent. The design question is the instrument that makes this cut.

Candidates considered. Five questions were formulated in the team and assessed against the criterion that a design question must be neither too broad nor too narrow (Table C.1). One was rejected as too broad (no place, no time, no intervention point), one as unanchored, and one as factually wrong — heat waves do not cause heat islands, which are a permanent consequence of urban surface properties and geometry (Oke 1982). The two survivors were a night-time removal question, well formed but discarding the team's central idea of *using* the heat, and a "turn heat into a resource" question whose seasonal-storage reading is neither verifiable within the project nor biologically supported: organisms store chemical energy across seasons, not sensible heat. The selected question keeps the night-time focus of the first and the resource framing of the second.

Design question (solution-neutral):

How might we let the heat that Zurich's existing buildings absorb during a summer day drive its own removal, so that the hotter it gets, the more the building cools itself?

The question names no solution — no coating, no greening, no geometry, no water — while fixing the location, the object (existing buildings, so retrofit rather than new build) and the season. What it adds to the two surviving candidates is the subordinate clause, and that clause is the content of the project: it demands a **self-regulating** response, in which cooling power rises with the heat load instead of being fixed. That is what separates the project from the incumbent measure. A bright surface (Section 2.3) has a fixed optical response: it reflects the same fraction at 20 °C as at 40 °C and does nothing once the sun has set. A self-regulating surface behaves like a biological one — transpiration in a leaf is driven by the same solar energy that heats the leaf, so the cooling effort scales with the load, without controller, sensor or pump. The clause also resolves the team's initial difficulty of carrying two systems of interest ("reduce heat" and "prevent heat"): these are two functions of one system, prevention acting in the day window and removal in the night window, coupled by the self-regulating link.

2.2 Context and constraints

Reference city and site. All context values, benchmarks and engineering checks refer to the City of Zurich (47.38° N, 8.54° E, about 400 m above sea level); fixing one city is what makes the constraints and KPI targets below checkable instead of generic. Zurich has an adopted heat-mitigation strategy, the *Fachplanung Hitzeminderung* (Stadt Zürich 2020a), with an implementation agenda (Stadt Zürich 2020c), a leaflet for building owners (Stadt Zürich 2021) and thematic maps (Stadt Zürich 2020b), so the project has a real client, a documented baseline and published target values instead of invented ones; open geodata (Stadt Zürich n.d.) and MeteoSwiss stations (MeteoSwiss n.d.) supply the boundary conditions (G, T_a, wind, humidity) for Section 3.2. As reference site we adopt the city's own model area *Modellierungsgebiet 03 — Geschlossene Randbebauung* (Stadt Zürich 2020a, pp. 148–151): two outward-closed blocks of five storeys enclosing courtyards in a dense central district, street space sealed from façade to façade. The city assesses this type as overheating comparatively quickly and already publishes modelled baseline and climate-optimised results for exactly this area — the methodical advantage — so our concept can be compared against a documented reference rather than our own assumptions. **[TODO: Record address, orientations, roof area and year of construction of the building you pick.]** The users are the residents of the upper floors, who carry the strongest load and usually have no mechanical cooling; the pedestrians in the canyon, who receive façade radiation at eye level; the owner, who pays for the retrofit; and the trades who install and maintain it.

When — the design case and its windows. The design case is not a single hot afternoon but a **multi-day heat wave**: five consecutive days with $T_{max} \geq 30\,°C$ and no rain. The concept relies on a stored resource — a system that performs on day one and is exhausted on day three has not solved the problem — and only a multi-day case puts the water reserve (F7) and the reset of the heat store (F4) under stress. KPI 6 follows from it. **[TODO: Confirm duration and frequency of such spells from MeteoSwiss records; state station and reference period.]** Within each day the design must perform in the three windows of Table 1, which follow one another without a gap because the surface never stops exchanging heat. The remaining hours, 06:00–11:00, are the margin the night window has to leave behind, so the state of the system at 06:00 is reported as the reset condition for the next day. The team's initial scoping put the second window at 20:00–24:00; it was moved to 18:00 and split from the night because a window must contain the instants at which its own KPIs are read (KPI 5 at 02:00), and because outdoor amenity and sleep quality need different KPIs.

Table 1: Critical operating windows within one day of the design heat wave. Sunset in Zurich in high summer is around 21:00, sunrise around 06:00.

Window	What dominates	Why it is critical	KPIs
Load 11:00–18:00	Absorbed short-wave radiation; clear-sky global irradiance $900\,W/m^2$ to $950\,W/m^2$ horizontal around solar noon, air up to $33\,°C$ to $36\,°C$.	The energy that has to be dealt with all night enters here; a sealed, sun-exposed surface can exceed $60\,°C$ (Stadt Zürich 2021). The city evaluates PET at 14:00, inside this window, so our results stay comparable to Table 2.	2, 3, 4
Evening 18:00–24:00	Low-angle sun on west-facing surfaces until sunset, then the start of the discharge into the canyon.	The window in which the effect is actually experienced: people are outdoors and then go to bed. West façades reach their highest incidence angle here, which is why the window starts at 18:00.	4, 5
Night 00:00–06:00	Long-wave emission to the sky and continued release of stored heat; clearest sky of the cycle.	Zurich's heat island reaches $7\,K$ at night against $1\,K$ to $2\,K$ by day (Stadt Zürich 2021) — the larger half of the problem, and the half the incumbent does not address. It sets a hard deadline: the store must be discharged by sunrise or it saturates over the following days.	1, 5, 6

Where on the envelope? A consequence of Zurich's latitude. Fixing the city fixes the solar geometry, and that already narrows the design space. At 47.38° N the solar altitude at noon on 21 June is $h = 90° - 47.38° + 23.44° \approx 66°$, so the direct beam strikes a horizontal roof at an incidence factor of $\sin h \approx 0.91$ but a vertical south façade at only $\cos h \approx 0.41$: in midsummer the flat roof receives more than twice the direct irradiance of a south façade at noon, and it has a far larger sky view factor for night-time emission. Both facts argue for the roof as a good place to *act* and against it as the place where the problem is. The heat a roof stores is released *upward*, to a sky it can see; the heat a façade stores is released *sideways*, into the canyon, at the height of the people in it. Since Section 1 fixes the pedestrian-level outdoor space as the place where the effect has to appear, the system of interest is the **façade** — not because it receives more radiation, but because its heat is released where it does harm. A façade concept also does not compete with photovoltaics for the roof area. The roof does not disappear; it changes role, from a second surface to be treated into the *interface with the good sky view*, which is what the concept in Section 3.2 requires. **[TODO: Confirm the load profile per orientation (Swiss Federal Office of Energy n.d.); east and west façades differ in the shape of the profile, not only its size.]**

Constraints. Table C.2 in Appendix C lists the context conditions in full and separates hard constraints (a violation disqualifies a concept) from soft ones (they weight the evaluation in Section 3.3). Seven are hard: **no auxiliary electrical energy** for the cooling function, **no potable water**, **retrofit only** onto an existing envelope, acceptability in the Zurich **townscape** including heritage protection, **no specular glare**, survival of the Swiss **annual cycle** to about −10 °C with freeze–thaw and UV, and **no appreciable increase in heating demand** — a high solar reflectance also reduces useful winter gains, and that trade-off must be quantified, not ignored. Glare and soiling are the two objections the city itself raises against bright envelopes (Stadt Zürich 2020a, p. 128). The soft constraints cover installability, maintenance, photovoltaic yield, dead load, embodied greenhouse gas and cost; the last two were originally KPIs, and Section 2.7 explains the move. Installation and maintenance are recorded as constraints rather than as subsystems, because a system of interest contains components, not people.

2.3 State of the art: what the City of Zurich already does

Scoping is incomplete without knowing what the client already does. Of the thirteen courses of action (*Handlungsansätze*) in the Fachplanung Hitzeminderung, four act on the building envelope: HA 09 green roofs, HA 10 green façades, HA 11 high-albedo façade and roof materials, and HA 12 shading of the outdoor space (Stadt Zürich 2020a, pp. 120–131). HA 11 is the direct incumbent of our design question, and the city already recommends it to building owners (Stadt Zürich 2021).

Table 2: Modelled effectiveness of the incumbent measures, from the Fachplanung Hitzeminderung. Day values are the physiologically equivalent temperature (PET) at 14:00, 2 m above ground; night values are air temperature. "Range" is the distance over which the effect is measurable.

Measure	Median	Max.	Range	Source
HA 11 bright façade — day (PET)	−1.0 K	−1.8 K	2 m to 3 m	(Stadt Zürich 2020a, Tab. 11, 128)
HA 11 bright façade — night	−0.1 K	−0.2 K	1 m to 2 m	
HA 11 bright roof — day (PET)	−1.2 K	−2.1 K	2 m to 4 m	
HA 11 bright roof — night	−0.1 K	−0.3 K	1 m to 2 m	
HA 06 high-albedo ground surface instead of asphalt — day (PET)	−1.5 K	−2.8 K	2 m to 4 m	(Stadt Zürich 2020a, Tab. 5, 114)
HA 06 — night	−0.2 K	−0.5 K	2 m to 3 m	

Baseline values this fixes for us. The city's model uses a solar reflectance of 0.3 for ungreened façades and 0.2 for asphalt, and classifies ≤ 0.30 as low and ≥ 0.80 as high albedo (Stadt

Zürich 2020a, pp. 114, 128–129). Our reference case and the target of KPI 2 are taken directly from these figures, so the comparison is made on the client's terms rather than on ours.

The gap — and the design opportunity. Table 2 contains the central finding for this project. The incumbent works during the day but is close to ineffective at night: −0.1 K median air temperature, against a heat island that reaches 7 K. The reason is structural, not a question of tuning. A bright surface reduces what is *absorbed*; it opens no path for releasing heat already stored in the construction, and its optical response is fixed — it does not do more when the load is higher, and nothing at all after sunset. A bio-inspired concept therefore earns its place if it does at least one of three things: open a removal path and so deliver a night-time effect; scale its response with the load, so the worst hours receive the strongest cooling; or resolve glare or soiling, the two constraints the city names for HA 11 and precisely the points where biological surfaces are strong.

Two precedents outside the city's own catalogue. The Fachplanung lists what the client does, not what the field has built. *StoColor Lotusan*, a micro-structured superhydrophobic façade paint modelled on the lotus leaf, has been on the market since 1999 and is stated by its manufacturer to resist algae and fungal growth *without biocide film protection* (Sto SE & Co. KGaA 1999) — a commercial answer to exactly the soiling weakness the city names, which establishes that the physical route to F10 is not speculative. *HydroCanopy*, inspired by frog mucous glands and by the density gradient of leaves in a canopy, combines solar regulation, ventilation, thermal inertia and evaporative cooling in one build-up (AskNature 2026e). It is the closest published prior art to our concept direction, which obliges us to state what we add: the self-regulating link between load and cooling power (F9), and an effect in the night window. **[TODO: For Section 5: p. 128 of the Fachplanung quotes a maximum PET reduction of 2.3 K in the running text but 1.8 K in Tab. 11 on the same page. Check the source and say which you use.]**

2.4 System view and project boundary

Figure 1 places the challenge in its system hierarchy. The hierarchy prevents tunnel vision; the boundary drawn afterwards prevents scope explosion.

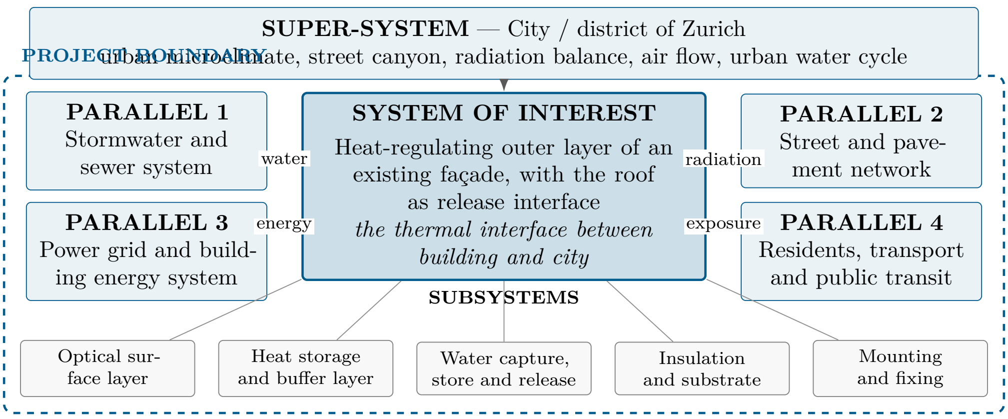

Figure 1: System view. The dashed line is the project boundary: everything inside it is designed and analysed by the team. The subsystems are components and functions, not people — the trades who install and maintain the system appear as constraints (Table C.2). What the comparison with each parallel system contributed is set out in Table C.3.

Table 3: Project boundary: what is deliberately in and out of the project.

IN the project	OUT of the project
Passive control of the short- and long-wave radiation balance of the envelope surface	Redesign of the building: structure, floor plan, windows, insulation thickness
The self-regulating link between heat load and cooling power	Any actively controlled system: sensors, valves, pumps, chillers, mechanical ventilation
Heat storage and delayed release within the outermost build-up, incl. the night window	Seasonal storage, boreholes, heat pumps, thermal networks, electricity generation
Capture, retention and metered release of rainwater within the build-up	Municipal water supply and sewage; street trees and district-scale greening
Mounting on an existing substrate, installability, durability of the function	Permitting, heritage approval, business model, full life cycle assessment

What the systems view changed. *The systems view changed our project scope because* it forced two corrections. First, we carried two systems of interest — reducing heat and preventing heat; the hierarchy showed these to be two *functions* of one system, coupled by the design question, so a single boundary can be drawn. Second, the comparison with the parallel systems (Table C.3) revealed the night sky and the discarded rainwater as the two heat sinks available without energy input, and the pedestrian in the canyon as the place where the effect must appear. Rainwater is currently drained away as fast as possible, yet under the "no potable water" constraint it is the only water available during a heat wave: seen from our system, a cooling reserve being discarded. This added F2 and F6–F8, the evening window, and KPI 5 and 6.

2.5 Functions and functional diagram

Functions are formulated as solution-neutral verbs answering "what must the design *do*?", and stated at the scale of the designed object rather than the city: the design regulates heat at the surface it occupies, and the city-scale effect is the *goal*, not the function — as the function of polar bear fur is to insulate, not to make the Arctic habitable.

Why F9 is physically plausible. Every removal path depends on surface temperature, but not equally strongly. Differentiated with respect to T_s, the self-regulation gradients are, in $\mathrm{W/(m^2\,K)}$, radiative $4\varepsilon\sigma T_s^3 \approx 6$, convective $h_c \approx 10$, and latent $\propto \mathrm{d}e_s/\mathrm{d}T_s$, which is steep because saturation vapour pressure roughly doubles per 10 K. The evaporative path is by far the steepest, which is why transpiration, not radiation, is the mechanism biology uses when a load must be tracked. **[TODO: Section 3.2: evaluate the three derivatives with your own assumptions; this check decides which mechanism carries F9.]**

2.6 Biologized questions

The functions are translated into questions to nature, which form the search strategy for Section 3.1. The challenge brief asks one question directly — what can be learned from desert plants and animals about passive cooling — and the answer shapes where we looked: the organisms most challenged by our problem, and unfazed by it, live where solar load is extreme and water is scarce. That is why the models of Section 3.1 are disproportionately desert organisms, and why the two that transfer best are those whose strategy is a structure rather than a metabolic process.

- How does nature use the heat it receives to get rid of that same heat?
- How does nature make a response stronger when the load is stronger, without a controller?
- How does nature release excess heat to the night sky?

Table 4: Function list. Main function, sub-functions and supporting functions.

ID	Function (verb)	Explanation / taxonomy link
Main function		
F0	Regulate heat gain and release at an urban surface	Modify / regulate physical state: manage temperature
Sub-functions — prevention (day window)		
F1	Reflect short-wave radiation	Prevent absorption of solar energy in the first place
F5	Shade a surface	Reduce the irradiated area, e.g. through surface geometry
Sub-functions — removal (both windows)		
F2	Emit long-wave radiation	Release heat to the sky, in particular in the atmospheric window
F3	Transfer heat to ambient air	Convective dissipation at the surface
F4	Store thermal energy temporarily	Buffer and shift the daily peak
Sub-functions — water path		
F6	Capture water	Collect rain or atmospheric moisture
F7	Retain water	Store it until it is needed during the heat peak
F8	Release water in a controlled way	Meter evaporation so the reserve lasts the whole heat wave
Sub-function — the core of the design question		
F9	Couple response strength to load	Increase removal (F2, F3, F8) as surface temperature rises, without external control
Supporting functions		
F10	Keep a surface clean	Maintain the optical function against soiling and algae
F11	Resist degradation	Withstand UV, frost, driving rain over the service life
F12	Attach to a substrate	Fix the build-up to an existing wall, installable by ordinary trades

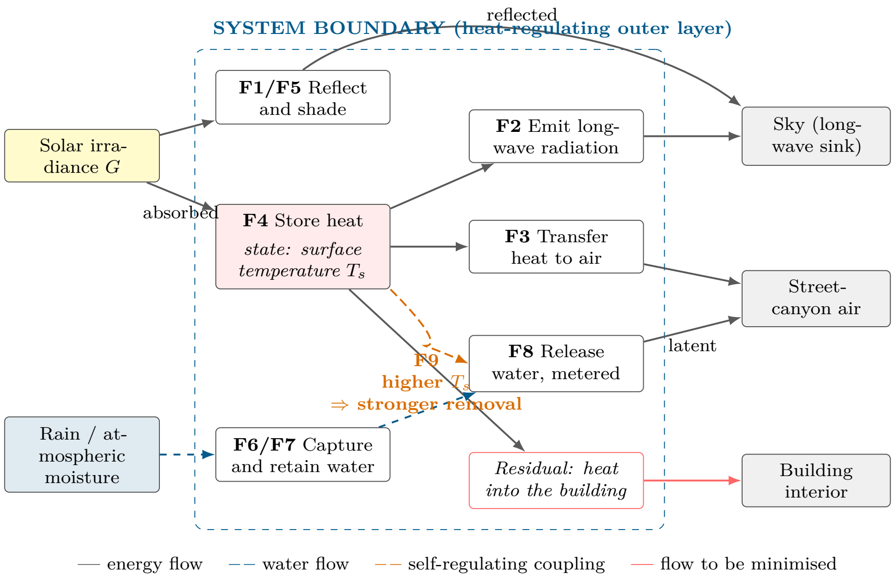

Figure 2: Solution-neutral functional diagram. Only functions and flows are shown, no components and no materials. The orange coupling is F9 and is what distinguishes this project from a fixed high-albedo surface: the state variable T_s feeds back into the strength of the removal functions. Supporting functions F10–F12 act on everything inside the boundary and are omitted for readability.

- How does nature reflect intense solar radiation without pigments that bleach, and without a mirror finish?
- How does nature capture, store and meter out scarce water so that a reserve lasts a whole dry period?
- How does nature buffer a daily temperature extreme instead of resisting it?
- How does nature keep a functional surface clean without cleaning it?

2.7 Performance criteria and KPIs

The KPIs are derived from the functions, interfaces and constraints identified above, not invented independently of them. They are used twice: to guide the engineering checks in Section 3.2 and to compare the concepts in Section 3.3.

Table 5: Performance criteria. Function → KPI → unit → target. Reference case: an existing Zurich façade with a solar reflectance of about 0.3, the value the city uses for un-greened façades (Stadt Zürich 2020a, p. 114).

#	Function → KPI	Unit	Target / benchmark
1	F9 — self-regulation gradient $\partial P_{\mathrm{cool}}/\partial T_s$	$\mathrm{W/(m^2\,K)}$	Clearly positive and larger than for a fixed high-albedo surface. **The defining criterion of this project.** **[TODO: set a target after the derivative check]**
2	F1 — solar reflectance α_{sol} (300–2500 nm)	–	≥ 0.80, the city's threshold for "high albedo" (Stadt Zürich 2020a, p. 128); baseline 0.3, ultra-white laboratory paints ≈ 0.98 (Li et al. 2021).
3	F0 — peak surface temperature reduction ΔT_s, clear sky, 14:00	K	≥ 15 K. **[TODO: confirm before claiming it]**
4	F0 — PET reduction at 2 m above ground, 14:00	K	Beat HA 11's −1.0 K median for a bright *façade*; the −1.2 K for a roof is context, not our benchmark (Stadt Zürich 2020a, Tab. 11). The metric the client uses.
5	F2, F4 — net cooling power at 02:00, clear sky; heat discharged 18:00–06:00	$\mathrm{W/m^2}$; $\mathrm{W\,h/m^2}$	$\geq 40\ \mathrm{W/m^2}$ instantaneous (Raman et al. 2014; Zhai et al. 2017), and a night-time air temperature effect better than HA 11's −0.1 K.
6	F6–F8 — potable water use; days of autonomy on harvested rainwater	$\mathrm{L/(m^2\,d)}$; d	Potable water = 0 (hard constraint); autonomy ≥ 5 days, so the reserve survives the design heat wave. **[TODO: derive the storage from 2.45 MJ/kg]**
7	F10/F11 — retention of the cooling function after soiling and weathering	% of initial	≥ 90 % after three years. Soiling is the weakness the city names (Stadt Zürich 2020a, p. 128), and aged cool roofs commonly lose a significant fraction of their reflectance (Santamouris 2014).

Two conditions on how these are read. First, the *reset condition*: KPI 1 and 5 are read per day, but over the five-day case the surface temperature and the state of charge of the store at 06:00 must not drift upwards from one day to the next. A concept that cools well on day one and saturates by day three fails the design case even if every single-day KPI is met. **[TODO: report the five-day trend, not only day one]** Second, KPI 3 measures a *surface* temperature and KPI 4 an effect on *people*: a high-reflectance surface can be 20 K cooler than a dark one while the effect on district air temperature is smaller by roughly two orders of magnitude (Santamouris 2014; Akbari et al. 2009). Conflating them is the most common error in urban-heat-island projects, so they are reported separately throughout. The targets are aspirational benchmarks set during scoping — hypotheses, not results — and each is confirmed, revised or declared unverified in Section 3.2. **[TODO: verify every cited benchmark against its primary source]**

Three changes to the team's original KPI list. Installed cost and embodied greenhouse gas were an eighth KPI; they are now constraints (Table C.2), because a KPI must follow from a *function* and neither does — which also brings the list to the seven the task asks for. *Urban heat island intensity* was dropped as a selection KPI, being a district-scale quantity a single retrofit cannot move measurably, and is replaced operationally by KPI 4. *Sustainability cost* was not measurable as stated and became the embodied-emissions constraint, which has a unit, a benchmark and a decision rule.

2.8 Relevant Life's Principles

The integrate step asks which principles are *relevant*, not how well the design achieves them; the assessment belongs to Section 4 and is made there against Table 14. Each is named here with the sub-principle claimed. **Be locally attuned and responsive** — *signal, antennae and response match* — is the principle the design question is built on, and F9 is it expressed as a function. **Be resource efficient** contributes *use low energy processes*, which is the hard energy constraint, and *one design meets more than one need*, a tie-breaker in Section 3.3. **Adapt to changing conditions** contributes *adapt to temporal change* (a five-day heat wave, not one afternoon), *withstand disturbance* (frost, driving rain, UV) and *maintain integrity through self-renewal* (KPI 7). **Use life-friendly chemistry** contributes *a safe subset* and *harmless breakdown products*, which is not decorative here: whatever leaches from the surface enters rainwater the concept evaporates at pedestrian level. **Integrate development with growth** is claimed only as *built to shape* — *creates more opportunities for life* is deliberately **not** claimed, because a permanently damp textured surface invites the algal growth KPI 7 penalises. And **evolve to survive** is **not claimed for the artefact** at all: a façade layer does not evolve across generations, so it is claimed for the *process* only.

The umbrella principle, *create conditions conducive to life*, requires a design to be *optimised rather than maximised*. That is a decision here, not a platitude: ultra-white coatings reach about 0.98 (Li et al. 2021), but KPI 2 targets the city's own ≥ 0.80, because further reflectance costs glare, winter gains and photovoltaic yield.

Quality check. System of interest clearly defined? Yes, and singular. Project boundary manageable? Yes, Table 3; seasonal storage and electricity generation are explicitly out. Functions solution-neutral? Yes, verbs only, with no components in Table 4 or Figure 2. Did the systems view reveal additional functions or constraints? Yes: F2 and F6–F8 from the sky and the discarded rainwater, the evening window from the residents, and the installation and maintenance constraints. Most important functions measurable? Yes, KPIs 1–7, each with a unit. Is the "How might we. . . ?" question open to different solutions? Yes: evaporative, radiative and geometric strategies all remain possible, and Section 3.2 must choose between them on evidence.

3 Bio-Inspired Design Process

The chain this section has to make visible is challenge → function → biology → abstraction → concept → engineering check → evaluation → iteration. Section 2 delivered the first two links. This section delivers the biology and the abstraction, and hands a set of transferable principles to the concept development in Section 3.2.

3.1 Discovering & Abstracting

3.1.1 Research path

Every search started from a verb in the function list of Table 4, never from an organism. This follows the rule that biological content is organised by the function it serves, which is also the structure of the Biomimicry Taxonomy and therefore of AskNature (Biomimicry Institute n.d.). Table 6 records which taxonomy branch was searched for which of our functions, and what it returned; the biologized questions themselves are in Section 2.6.

Table 6: Research path. Each of our functions was translated into a taxonomy branch before any organism was considered.

Function	Taxonomy branch searched	Models it returned
F1, F5	Modify → physical state → light/colour; Protect from non-living threats → light	Desert snail, *Morpho*, *Cyphochilus*, tree bark, cactus ribs
F2, F3	Transform/convert energy → thermal, radiant; Distribute → energy	Elephant ears, elephant hairs, jackrabbit ears, Saharan silver ant
F4	Store → energy	Sea star, desert snail
F6–F8	Capture, absorb or filter → liquids; Store → liquids; Distribute → liquids	Leaf stomata, camel fur, honeybee water collection, bryophytes
F9	Sense signals → temperature *paired with* Process signals → respond to signals	Honeybee fanning thresholds, leaf stomata, pine cone, elephant hot spots
F10, F11	Protect from non-living threats → dirt/solids, light, ice	Lichen, lotus leaf

F9 is worth noting: it is the only one of our functions that does not sit in a single branch of the taxonomy. It is the *pairing* of "sense signals: temperature" with "process signals: respond to signals", which is the same construction the Life's Principles checklist uses when it asks that "the signal it sends, the antennae it uses to detect the signal, and its response match" (Baumeister et al. 2014). That the central function of our design question has to be assembled from two branches is itself a finding: it is why searching by organism would not have produced it.

Sources, in the order they were used. AskNature strategies and innovations to find candidates (Biomimicry Institute n.d.); the *Genius of Biome* report for the temperate broadleaf forest, which is the biome Zurich lies in, for models already translated to the built environment (Biomimicry 3.8 and HOK 2013); primary literature for the mechanisms; and commercial products where the transfer has already been made, which are treated as state of the art rather than as models. AskNature and AI-assisted search were used to *find* candidates only. Every mechanism claim carried into this report is referenced — but not yet uniformly to a primary paper: twelve of the twenty entries in Table 7 currently rest on the AskNature strategy page alone. That is a stated shortfall rather than a claimed standard, and the open item below names the two papers already identified as replacements.

The inverse question. The method also asks what nature would *not* do, and the answers narrowed the search as effectively as the positive ones. Nature would not use a mirror finish, would not run a compressor, would not pump drinking water onto a wall, and would not build one fixed response for both January and July. Three of those four are constraints in Table C.2; the fourth is the seasonal problem this project does not yet solve.

Which lenses were applied. Of the five lenses in the method, two did most of the work. The *operating-condition lens* — intense solar radiation, surfaces above 60 °C, freeze–thaw, weeks without rain — led to the desert organisms. The *local lens* led to (Biomimicry 3.8 and HOK

2013), because Zurich sits in the temperate broadleaf forest biome and that report translates exactly this biome into building design. The *function lens* produced Table 6. The naturalist and ecological lenses contributed context but no model that survived selection.

3.1.2 Models found, and the selection decision

Table 7: Biological models found, and the selection or de-selection decision. The criterion applied throughout: a model must be transferable to a *passive surface* that survives twenty years of the Swiss annual cycle.

Organism / model	**Function**	**Mechanism**	**Decision**
Prevention — reduce what is absorbed			
Desert snail *Sphincterochila boissieri*	F1, F4	Shell reflects ≈ 90 % of visible and ≈ 95 % of near-infrared light; in extreme heat the aperture is sealed with a calcified plate and the animal withdraws, leaving an insulating air space between body and hot ground (AskNature 2026k)	**Selected.** Passive, mineral, non-metabolic — the closest analogue to an envelope we found
Morpho butterfly	F1	Transparent chitin–air layers a few hundred nanometres apart diffract and recombine light; colour comes from geometry, and the disordered structure keeps the reflection diffuse rather than specular (AskNature 2026c)	**Selected, with a stated limitation.** It decouples appearance from absorptance, which is what the townscape constraint requires — but the *Morpho* demonstrates structural *colour*, not near-infrared heat rejection. That transfer needs separate technical verification; the organism does not establish it
White beetle *Cyphochilus*	F1	Brilliant whiteness from a disordered chitin network only 7 ± 1.5 µm thick, without pigment (Vukusic et al. 2007)	**Selected** as the achromatic case of the same principle
Saharan silver ant *Cataglyphis bombycina*	F1, F2	Triangular hairs scatter visible and near-infrared light and simultaneously raise emissivity in the mid-infrared (Shi et al. 2015)	**Selected.** The only model that does F1 and F2 in one structure
Cactus ribs	F5, F3, F2	Peaks shade the troughs; the cooler trough air absorbs heat and rises out, while the folds enlarge the emitting area and disturb the airflow around the body (AskNature 2026d)	**Selected.** Geometry only — no material, no water, nothing to bleach

Organism / model	Function	Mechanism	Decision
Tree bark	F1, F2	Reflects part of the incoming radiation while emitting efficiently; rough layering shades itself and traps insulating air (AskNature 2026a)	**Selected.** A dead outer layer protecting a living structure — the same role our build-up has
Removal — get heat out again			
Elephant ears	F2, F3	Warm blood is carried from the core into large thin ears; the surface releases it, and flapping raises the airflow (AskNature 2026h)	**Selected.** Separates the place of absorption from the place of release
Elephant skin hairs	F3	At this density the hairs do not insulate — it lies well below the crossover at which insulation becomes dominant — so each acts as a small heat-transfer *fin*, reaching through the still boundary layer that clings to the skin so that moving air reaches the surface. Reported additional heat loss up to 23 % (University of Notre Dame 2021)	**Selected.** Directly relevant at the low wind speeds of a canyon
Jackrabbit ears	F9, F2	Vasodilation varies the blood flow to a thin radiating surface; at $\approx 30\,^{\circ}\mathrm{C}$ air temperature the ears shed all excess heat, and the strategy exists to avoid spending water (AskNature 2026g)	**Selected.** The water-free route to a load-coupled response
Elephant skin hot spots	F9	Vasodilation in scattered patches rather than uniformly; the number of open “thermal windows” varies with the load (AskNature 2026f)	**Selected.** Regulation by active *area* rather than by material
Water path and storage			
Leaf stomata	F9, F8	Environmental and chemical signals change the solute concentration in the two guard cells; osmosis moves water accordingly, turgor changes, and the cells bend open or shut the pore. The mechanism therefore regulates gas exchange *and* limits water loss in the same movement. Evaporation itself pulls the water up, with no pump; leaf surfaces cool by 11 K to 17 K (Biomimicry 3.8 and HOK 2013, pp. 62–65)	**Selected.** The clearest natural instance of F9
Camel fur over sweat glands	F7, F8	The fur works on all three transfer paths at once: it limits radiative gain through its light colour, conduction through the air trapped between the hairs, and convection by damping air movement close to the skin — and only then is sweat spent on what is left. In the technical reproduction of that arrangement the pair held a sample 7 K below ambient for 200 h, five times longer than the wet layer alone (AskNature 2026i)	**Selected.** Turns the water reserve from a quantity into a duration
Sea star	F4, F7, F9	Takes up more cold water at high tide after a hot low tide, so the previous load sets the charge; fails once the sea is no longer colder than the air (AskNature 2026b)	**Selected.** Charge-before-load, including its own failure mode

Organism / model	Function	Mechanism	Decision
Bryophytes	F6, F7	No cuticle, no roots; a living layer over a decaying one holds moisture behind a still-air boundary, and the plant survives drying out (Biomimicry 3.8 and HOK 2013, pp. 53–56)	**Not carried** into the concepts — superseded by the camel bilayer, which achieves the same with a documented duration
Self-regulation and supporting functions			
Honeybee fanning thresholds	F9	Each worker begins fanning at a slightly different temperature, and the spread is genetic; the number of active bees rises with the load, so binary units produce a graded colony response (AskNature 2026l)	**Selected.** Resolves whether a threshold can satisfy F9
Pine cone scales	F5, F9	Two bonded layers swell differently with moisture: taking up moisture makes one expand more, so the scale bends and closes, and as it dries it returns towards the open state. The movement is reversible and the cone is dead tissue, moved entirely by the weather (AskNature 2026j)	**Selected.** A passive actuator with no energy input
Lichen *Lecanora conizaeoides*	F10, F11	Roughness at several scales keeps droplets perched on the peaks so they are shed, while the hydrophobic micropores between them stay dry and remain open to gas exchange even in the wet: the surface separates liquid transport from gas transport across the same plane (Biomimicry 3.8 and HOK 2013, pp. 41–44)	**Selected.** A physical answer to soiling, not a chemical one
Desert shrub *Encelia farinosa*	F1, F5	Dense leaf pubescence raises reflectance and is adjusted seasonally (Ehleringer and Mooney 1978)	**Selected** as the biological precedent for a seasonally switching surface
African elephant skin cracks	F6–F8	A network of micrometre-scale cracks retains water and extends evaporative cooling (Martins et al. 2018)	**Selected.** Third mechanism from the same organism
De-selected, with reasons			
Termite mound	F3	The classic reference, but the “chimney air-conditioning” account is contested; measurements support wind-driven transient gas exchange rather than a thermosiphon (Turner and Soar 2008)	**De-selected.** The evidence does not support the mechanism, and the building usually cited for it uses fans
Namib desert beetle	F6	Hydrophilic–hydrophobic patterning harvests water from fog	**De-selected.** Wrong season for our climate: fog in Zurich is a winter phenomenon, and a heat wave brings neither rain nor fog

Organism / model	Function	Mechanism	Decision
Thomson's gazelle carotid rete	F3	Counter-current heat exchange cools the brain	**De-selected** as a whole-organism model — internal physiology. The counter-current *arrangement* is retained as an option for the liquid loop in Section 3.2
Tenrec aestivation	—	Dormancy through the hot season	**De-selected.** A behaviour; an envelope has none
Green façades and roofs	F1, F5	Shading and evapotranspiration from planting	**Out of scope** by the project boundary (Table 3), and in direct competition with our own water reserve: planting needs most water exactly when least is available

Two features of Table 7 are deliberate. First, the same organism appears three times — the elephant supplies ear-based transport, boundary-layer hairs and localised hot spots, and these are three different mechanisms serving three different functions, not one model counted three times. Second, five models were rejected *with a stated criterion* rather than quietly dropped. The criterion is the one at the head of the table, and it is worth stating explicitly because it explains an asymmetry in what survived: most cooling in nature is metabolic or behavioural, and an envelope has neither metabolism nor behaviour. The two models that transfer most directly — the snail shell and the cactus rib — are the two that are *structures rather than processes.*

[TODO: Before hand-in: open every AskNature page cited in this table and replace it with the primary paper wherever one exists. Two are already identified: the silver-ant film reproduction is *Biologically inspired flexible photonic films for efficient passive radiative cooling*, PNAS 2020, and the adaptive infrared material is *A dynamic thermoregulatory material inspired by squid skin*, Nature Communications 2019. The course rule is explicit that AskNature finds candidates and the paper is the evidence.]

3.1.3 Mechanisms in depth

Three mechanisms are developed further here, one per functional family. They were chosen because each answers a question Section 2 left open.

Prevention — why reflection alone is not the strategy. The desert snail survives on sun-exposed ground at up to 50 °C without moving, without water, and without a metabolism that could help. Its shell reflects about 90 % of visible and about 95 % of near-infrared light, which is above the $\alpha_{\mathrm{sol}} \geq 0.80$ that KPI 2 asks for and close to the ultra-white laboratory paints of Li et al. (2021) — achieved with calcium carbonate rather than a manufactured pigment. The part that matters for us is what happens next. In extreme conditions the animal seals the aperture and retracts into the second whorl, so an air space separates its tissue from the shell wall nearest the hot ground. Reflection reduces what is absorbed; the air space stops what is still absorbed from reaching what must be protected. For a retrofit this maps onto a construction that already exists: a reflective outer skin in front of a ventilated cavity. That matters for the night window, because the 7 K the city loses at night is heat released from mass that was charged during the day, and a decoupled outer skin charges far less of it.

Regulation — how a switch can behave like a controller. In a honeybee colony the temperature at which an individual worker begins to fan varies between individuals, and the variation is genetic: a colony has one queen but many fathers, so it contains patrilines with different thresholds. As the nest warms, first a few bees fan, then more, then many, and the nest is held below about 36 °C with no central control at all. Each bee is a binary switch; the colony response is continuous. This resolves an objection that would otherwise disqualify every switching material considered in Section 3.2: a single threshold gives an almost infinite $\partial P/\partial T_s$ at the switching point and roughly zero either side of it, which is not the graded response KPI 1 asks for. A *distribution* of thresholds across many small units restores it. The same pattern appears in Biomimicry 3.8 and HOK (2013) as the "lots of littles" principle, arrived at independently.

Water — why metering matters more than harvesting. A leaf cools itself by 11 K to 17 K through pores whose two guard cells swell and shrink with their water content, opening and closing the pore as they do; the water is drawn up by the evaporation itself, with no pump anywhere (Biomimicry 3.8 and HOK 2013, pp. 62–65). The valve responds directly to heat and dryness, so the effort tracks the load. The camel supplies the complementary half. Its fur is insulation worn *in* the heat, and in the technical reproduction of that arrangement — an evaporating hydrogel underneath a heat-blocking but vapour-permeable aerogel — a sample was held 7 K below ambient for 200 h, five times longer than the wet layer alone (AskNature 2026i). Insulating the wet layer, rather than exposing it, is what converts a fixed volume of water into a duration. Since KPI 6 asks for five days of autonomy and Zurich supplies no rain during the design heat wave, metering is the binding problem and harvesting is not.

[TODO: Add one figure per mechanism above. Prompts for generating them are in `Working documents/image_prompts.md`; a generated figure is an illustration and must be captioned as schematic, and the AI use belongs in Appendix A.]

3.1.4 Abstraction into design principles

Table 8 keeps the three columns strictly separate, as the method requires: what was observed, what the observation abstracts to, and what it might become technically. Only the third column contains anything that could be built, and nothing in it has been selected yet — that is the work of Section 3.2.

One principle, found three times. The jackrabbit and the elephant regulate by how much *area* is active, the bees by how many *units* have switched, and a bilayer shading element by how far it has *deployed*. All three modulate how many elements act rather than how strongly each one acts, and all three arrive at a graded response without a controller. That convergence from three unrelated organisms is the strongest abstraction this project has produced, and it is the one carried into every concept in Section 3.2.

3.1.5 Life's Principles evident in the selected strategies

The principles below are the ones the *selected models* actually exhibit; Section 4 assesses how well our final concept applies them, which is a different question.

Table 8: From biological observation to transferable design principle. The third column lists possibilities, not decisions.

Biological observation	Abstracted principle	Possible technical interpretation
The snail's shell reflects strongly *and* an air space separates it from the hot substrate	Reflect first, then decouple whatever still gets through with a layer of air	Reflective outer skin over a ventilated cavity; structural white instead of a pigment
Morpho colour comes from layer geometry, not pigment, and the disordered structure stays diffuse	Separate what a surface looks like from what it does with heat	Structurally coloured or near-infrared-selective finish that keeps the existing façade colour
Cactus peaks shade the troughs; cooler trough air absorbs heat and rises out	Use folded geometry for self-shading, more emitting area and better airflow	Ribbed or profiled façade and paving relief
Sparse hairs reach through the still air layer clinging to the skin	Break the boundary layer so moving air reaches the surface	Millimetre-scale texture on façade and pavement
The jackrabbit varies blood flow to a thin radiator; the elephant opens patches	Regulate by varying the *thermal connection* between the store and a radiating surface, and place that radiator only where conditions favour loss	Pumpless density-driven liquid loop from the canyon face to a parapet-level radiator with a good sky view
Guard cells open and close the pore in response to heat and dryness	Build the control into the opening, so the escape route widens with the load	Load-driven valve or wick, rather than a controlled one
Camel fur insulates *over* the evaporating skin	Cut the load first, then spend a little water on the remainder, and cover the wet layer so the reserve lasts	Absorbent layer beneath a vapour-permeable cover
Worker bees each switch at a different temperature	Spread many binary elements across a range of trigger points and the sum responds continuously	Any switching element built as many small units with a deliberate spread of switching temperatures
Pine cone scales bend because two bonded layers swell differently	Make moving parts from two layers that respond differently and let the environment actuate them	Bilayer shading element; hygroscopic or thermal actuation, no motor
The sea star charges on cold water sized by the previous hot exposure	Charge the store during the cool phase, sized by the load of the previous cycle	Night-charged store, with an explicit reset condition across the five-day case
Lichen roughness at several scales sheds water while staying breathable	Use geometry rather than chemistry to shed water and stay clean	Multi-scale textured finish; no biocide in the film

- **Be locally attuned and responsive** — present in every model that couples its response to the local load, and named by the checklist in the form our F9 requires: the signal, the antenna that detects it and the response must match (Baumeister et al. 2014).
- **Be resource efficient (material and energy)**, sub-principle *use low energy processes* — none of the selected mechanisms consumes energy; the load itself drives them.
- **Adapt to changing conditions**, sub-principles *incorporate diversity* and *maintain integrity through self-renewal* — *Encelia* and the pine cone for the first, the lichen for the second.
- **Use life-friendly chemistry** — *Cyphochilus* achieves with geometry what industry achieves with titanium dioxide. The lichen is the more careful case: its water repellency does involve a compound, but that compound works *with* a multi-scale geometry rather than instead of it, and nothing is released to do the job.

[TODO: Section 4 must *assess* these, not list them again. The rubric marks down a list.]

3.2 Creating: Engineering Concept Development

3.2.1 Ideation

[TODO: Run and document Crazy 8s: eight technically different ways of transferring the abstracted principles of Table 8. Vary the mechanism, not the shape, and do not evaluate while generating. Include a photo or scan of the sheets. Warning: four of the eight ideas in the lecture's own worked example (night flush, evaporative inlet, thermal buffer, shading skin) overlap our function list. If our eight reproduce that list we have copied the example instead of using our own abstractions.]

3.2.2 Morphological matrix

The method is function-based concept generation (Nagel et al. 2010): the rows of the matrix are functions, the columns are alternative principles that serve them, and a concept is a path across the rows. The rows are chosen, not inherited. Of our thirteen functions, a row belongs in the matrix only where alternative principles would lead to a meaningfully different concept. On that criterion F0 is excluded as the main function, F10 and F11 are requirements and engineering checks rather than sources of concept variants, and F12 is included only if the mounting really changes the architecture. That leaves five rows.

[TODO: Read at least three coherent combinations off this matrix and name them. A column is not a concept: check that the options combined are physically compatible. Then select two that are technically distinct — different physics, not different styling — and develop them below.]

3.2.3 Concept A — [TODO: name]

[TODO: Annotated drawing plus description: components, functional chain, flows, candidate materials, key technical parameters. Annotate inlet, channels, driving effect, outlet, heat path, materials, key dimensions and expected performance, and state explicitly which parts are inspired by biology and which are conventional engineering.]

3.2.4 Concept B — [TODO: name]

[TODO: Must be technically distinct from A. The four directions currently on the table are: a switching coating (prevention, no night effect); a self-deploying shade

Table 9: Morphological matrix. Rows are the five functions for which alternative principles produce genuinely different concepts; columns are the technical interpretations abstracted in Table 8.

Function	**Option A**	**Option B**	**Option C**
F1, F5 Reduce heat input	Structural white or near-infrared-selective finish	Ribbed, self-shading geometry	Deployable shading element
F2, F3 Move heat away	Broadband emitter with a good sky view	Boundary-layer texture for convection	Density-driven liquid loop to a parapet radiator
F4 Store and buffer	Thermal mass in the build-up	Latent store (phase change)	No added storage; decoupling only
F6–F8 Handle water	Absorbent layer under a vapour-permeable cover	Wick with load-driven release	No water path
F9 Couple response to load	Switching material, many units, spread thresholds	Load-driven valve or pore	Bilayer element that deploys with heat or moisture

(prevention plus an open night sky, acts at pedestrian level); an evaporative build-up with a water path (removal, metered per the camel bilayer); and a ribbed geometry (prevention and convection, no material and no water). The pairing with the widest physical separation is the deploying shade against the evaporative build-up.]

3.2.5 Engineering Canvas and critical questions

A critical question names a quantity, a threshold and a scale. Three follow directly from the abstractions of Section 3.1 and are carried forward as candidates:

1. Is the buoyancy in a pumpless density-driven loop sufficient to move the required heat flux over the height available in a five-storey block, and does the loop survive a Swiss winter?
2. At what wind speed must a self-deploying shading element furl, and what furls it? A thermal actuator cannot sense wind, so an independent and equally passive release is required before the concept is safe over a street.
3. Can a switching element be produced with a deliberate *spread* of switching temperatures, as the honeybee threshold model requires, or does the chemistry force a single sharp threshold?
4. **[TODO: Select two or three of these, or replace them once the concepts are fixed. They must be the assumptions that could change the design decision.]**

3.2.6 Engineering check 1: steady-state surface energy balance

[TODO: This check is prepared because it is the most direct way to test KPI 1 and KPI 3. Fill in your own numbers, state every assumption, and keep the units.]

For an opaque surface in steady state, the energy balance per unit area is

$$(1-\alpha_{\mathrm{sol}})\,G \;+\; \varepsilon L_{\downarrow} \;-\; \varepsilon\sigma T_s^4 \;-\; h_c\,(T_s - T_a) \;-\; q_{\mathrm{cond}} \;=\; 0, \tag{1}$$

where α_{sol} is the solar reflectance, G the global irradiance on the surface, ε the thermal emittance, $L_{\downarrow}$ the incoming long-wave radiation from the sky, $\sigma = 5.67 \times 10^{-8}\,\mathrm{W/(m^2\,K^4)}$ the Stefan–Boltzmann constant, T_s the surface temperature, T_a the air temperature, h_c the convective heat transfer coefficient and q_{cond} the heat conducted into the construction.

Table 10: Engineering Canvas. One per concept; the output of the canvas is the list of critical questions below it.

Field	**Concept [TODO: A / B]**
Function — what must it achieve?	**[TODO:]**
Biological mechanism	**[TODO:]**
Abstracted principle	**[TODO:]**
Technical principle — what is transferred?	**[TODO:]**
Architecture — components and interfaces	**[TODO:]**
Inputs / outputs — material, energy, information	**[TODO:]**
Requirements / KPIs	**[TODO:]**
Assumptions	**[TODO:]**
Uncertainties — what could invalidate it?	**[TODO:]**
Next check — what should we calculate or test?	**[TODO:]**

Table 11: Assumptions for the energy balance. Every value needs a source.

Quantity	**Value used**	**Justification / source**
G	**[TODO:]** $\mathrm{W/m^2}$	**[TODO: MeteoSwiss, clear-sky summer day; Table 1 gives 900** $\mathrm{W/m^2}$ **to 950** $\mathrm{W/m^2}$ **horizontal]**
T_a	**[TODO:]** $^\circ\mathrm{C}$	**[TODO: Table 1: 33** $^\circ\mathrm{C}$ **to 36** $^\circ\mathrm{C}$ **in the load window]**
h_c	**[TODO:]** $\mathrm{W/(m^2\,K)}$	**[TODO: low wind speed in a street canyon]**
$L_\downarrow$	**[TODO:]** $\mathrm{W/m^2}$	**[TODO: clear-sky estimate; state the correlation used]**
α_{sol} reference	0.3	Value the city uses for un-greened façades (Stadt Zürich 2020a, p. 114)
α_{sol} concept	**[TODO:]**	**[TODO: target** ≥ 0.80**, KPI 2]**
ε	**[TODO:]**	**[TODO: state whether the concept changes emittance or only reflectance — these are separate questions]**

Result and meaning. **[TODO: Solve Eq. (1) for T_s for the reference and for the concept, report both, and state explicitly what the difference does and does *not* prove (see the note on KPI 3 in Section 2.7).]**

3.2.7 Engineering check 2: [TODO: title]

[TODO: Options that are realistic within the week: (a) a simple outdoor measurement — identical plates with different surfaces, logged over a few hours; (b) a transient 1-D model of the build-up over a diurnal cycle; (c) a water balance — litres of rainwater per square metre per hot day against the latent heat of vaporisation of 2.45 MJ/kg, checked against the 200 h the camel bilayer achieved (AskNature 2026i); (d) a material and cost comparison. Report method, raw data, result and uncertainty.]

A scale check that applies to four of our models. Micro-scale effects do not automatically survive being made square metres large. The *Cyphochilus* whiteness, the *Morpho* colour, the lichen roughness and the stomatal valve all live at micrometre scale, and the failure mode is named explicitly in the method: upscaling can break the function. **[TODO: State for each of the four whether the effect scales, and cite the evidence. For the lotus effect the answer is partly known, because a lotus-effect façade paint has been manufactured since 1999 (Sto SE & Co. KGaA 1999).]**

3.3 Concept Evaluation & Iteration

The last three rows are not KPIs and are deliberately separate from them. They carry the questions a KPI table cannot answer: whether the concept can be built, whether it genuinely transfers the biology or merely resembles it, and how well supported the ratings above are. The third of these is the one that prevents a confident-looking table from concealing guesswork.

3.3.1 Review and feedback

[TODO: World café peer feedback, instructor feedback, and AI used as a critical reviewer. For every AI comment record the decision — accept, reject or investigate — and the justification. A documented rejection is worth more than a list of accepted comments, because it shows judgement. This belongs in Appendix A as well.]

3.3.2 Selection and Final Concept V2

[TODO: State which concept, or which justified hybrid, was selected, and describe V2 in detail. The minimum evidence of iteration is one named weakness or uncertainty in the earlier concept and a documented response to it in V2.]

3.3.3 Remaining uncertainties

[TODO: What is still unproven, and what should be tested or developed next. The scale question above and the wind release for a deploying element are the two that are already known to be open.]

Table 12: Decision matrix. Ratings 1–5, weights from Section 2.7. Every rating must be justified in the text by a calculation or a reasoned estimate.

Criterion	Weight	Concept A	Concept B	Evid used
KPI 1 – Self-regulation gradient	[TODO:]	[TODO:]	[TODO:]	[TO]
KPI 2 – Solar reflectance	[TODO:]	[TODO:]	[TODO:]	[TO]
KPI 3 – Peak surface temperature	[TODO:]	[TODO:]	[TODO:]	[TO]
KPI 4 – PET at 2 m	[TODO:]	[TODO:]	[TODO:]	[TO]
KPI 5 – Night radiative cooling	[TODO:]	[TODO:]	[TODO:]	[TO]
KPI 6 – Water path and autonomy	[TODO:]	[TODO:]	[TODO:]	[TO]
KPI 7 – Durability of the function	[TODO:]	[TODO:]	[TODO:]	[TO]
Feasibility	[TODO:]	[TODO:]	[TODO:]	[TO]
Quality of the biomimetic transfer	[TODO:]	[TODO:]	[TODO:]	[TO]
Evidence and uncertainty	[TODO:]	[TODO:]	[TODO:]	[TO]
Life's Principles fit	[TODO:]	[TODO:]	[TODO: use the assessment in Table 14]	
Weighted total		[TODO:]	[TODO:]	

Table 13: Final output of the creating and evaluating phase: Concept V1a/b → engineering review → evidence → Concept V2.

What changed	Why it changed	Evidence that informed the change
[TODO:]	[TODO:]	[TODO:]
[TODO:]	[TODO:]	[TODO:]

4 Sustainability & Life's Principles

4.1 What this section assesses, and what it does not claim

Section 2.8 named the principles this project works under. Naming them is not applying them, and the difference is the whole content of this section: each principle below is stated as a *claim*, followed by the evidence for it and by the point at which the concept falls short of it. A principle with no shortfall named is a principle that has not been assessed.

Two limits are stated at the outset rather than buried. First, the assessment of the final concept can only be as specific as the concept, and Section 3.3 has not yet fixed it; the parts of this section that depend on that choice are marked. Second, no life cycle assessment was carried out. Where embodied emissions are discussed they are discussed as a *criterion with a decision rule*, not as a computed result, and the words "sustainable" and "resource efficient" are avoided as descriptions of the concept because on the present evidence neither could be defended.

4.2 Life's Principles applied to the concept

4.3 Eco-efficient or eco-effective?

Cradle to Cradle draws a distinction that this project has to face directly (Braungart and McDonough 2002). An *eco-efficient* design is less bad: it does the same thing while emitting less. An *eco-effective* design is actively good: its outputs are inputs somewhere else. The standard illustration is a brake pad — one version emits fewer particles but still emits thousands of tonnes, the other is made of a material that can safely re-enter a biological cycle. Nature, the argument runs, is not efficient at all: a cherry tree produces blossom wastefully, and is nonetheless effective because every part of that waste is a nutrient.

Measured against that distinction, most of this concept is **eco-efficient and no more**. A façade that absorbs less solar radiation is a less bad façade; it produces no nutrient for anything. Saying so plainly is more useful than claiming otherwise, because it locates the one place where the project does better.

That place is the water path. The rainwater this concept collects is water the city currently drains away as fast as possible — Section 2.4 identifies it as a cooling reserve being discarded, and the storm sewer is the parallel system that discards it. Taking that output and making it the input of the cooling function is *waste equals food* in the strict sense, and it is the only claim of this kind the project can make honestly. It is also an argument the comparison in Section 3.3 has to weigh: *if* the wet direction is carried forward, it is on this ground and not on cooling power alone, because a dry concept is merely efficient while the wet one closes a loop that is currently open.

4.4 The material perspective: which layer belongs in which cycle

Cradle to Cradle divides materials into *biological nutrients*, which are consumed and must return safely to the environment, and *technical nutrients*, which are used without being consumed and are recovered (Braungart and McDonough 2002). The rule that assigns a material to one or the other is about wear:

> Anything that is consumed — worn out — has to come from the biological cycle. Anything that is merely used, without wear, can come from the technical nutrient cycle.

Applied layer by layer, this decides the material strategy of the build-up before any material has been chosen, and it produces one uncomfortable result.

Table 14: Life's Principles assessed against the concept. The last column is the part that matters: a principle claimed without a shortfall has not been assessed.

Principle and sub-principle claimed	How the concept applies it	Where it falls short
Be locally attuned and responsive *the signal, the antenna and the response match*	F9 is this principle expressed as a function: the surface temperature is the signal, the physics of the removal path is the antenna, and the cooling power is the response. Both sinks used — the night sky and rainwater the city currently drains away — are local and freely available.	The response is attuned to *temperature*, not to the quantity that actually matters to people, which is radiant load at eye level. The concept optimises what it can sense, not what it is for. **[TODO: Revisit once V2 is fixed.]**
Be resource efficient *use low energy processes* and *one design meets more than one need*	The cooling function consumes no auxiliary energy; this is a hard constraint, not an aspiration. Multi-functionality is set as an explicit tie-breaker for Section 3.3: a build-up in which one layer carries several functions is to be preferred to one that needs a layer per function.	Nothing has yet been counted. "Resource efficient" is a claim we cannot make until the embodied emissions of the added build-up are estimated against the cooling it delivers over its service life.
Adapt to changing conditions *adapt to temporal and spatial changes, maintain integrity through self-renewal*	The design case is a five-day heat wave with a reset condition, not one hot afternoon. Durability against soiling is a criterion in its own right (KPI 7) rather than an afterthought.	The seasonal half of the principle is weakly served. A surface optimised for July is a surface that also faces January, and the winter trade-off is currently a constraint we accept rather than a behaviour we design.
Use life-friendly chemistry *safe subset, harmless breakdown products*	The decision that self-cleaning must be physical rather than chemical follows directly from this principle, and it is a design constraint, not a preference: anything that leaches from the surface enters rainwater which the concept then evaporates at pedestrian level.	The principle is currently satisfied by *exclusion* — we have ruled substances out. No substance we intend to use has yet been checked in a toxicity database. See Section 4.7.
Integrate development with growth *built to shape*	A modular element sized to a façade grid produces no offcuts. On-site material waste in construction is put at 10 % to 15 % by the Ellen MacArthur Foundation (Ellen MacArthur Foundation 2015), so this is a measurable quantity rather than a gesture.	Claimed narrowly and on purpose. The sub-principle *creates more opportunities for life* is **not** claimed: a permanently damp textured surface invites exactly the algal growth KPI 7 penalises.
Evolve to survive	**Not claimed for the artefact.** A façade layer does not evolve across generations. It is claimed for the *process*: this report builds on the incumbent measure rather than starting from zero, and documents what was de-selected and why.	—

Table 15: The wear rule applied to the build-up.

Layer	Worn or used?	Consequence
Outer finish or coating	**Worn** — UV, driving rain, freeze–thaw, soiling	Must be biologically benign in breakdown. It is abraded into the street, and into the run-off we collect
Water-carrying absorbent layer	Worn, and in direct contact with the run-off	Same requirement, for the same reason
Ventilated cavity, sub-structure, fixings	**Used**, not consumed	Technical cycle: recoverable, demountable, and worth specifying as separable single materials
Masonry behind	Untouched by the retrofit	Outside the product

The uncomfortable result. The layer we would most readily engineer with chemistry — the outer finish, where the optical performance lives — is precisely the layer that must be biologically benign, because it is the one that wears away into the place we are trying to improve. That inverts the usual order of a coating specification, in which durability and optical performance are chosen first and the environmental profile is checked afterwards.

4.5 Circular design principles: an honest status

Table 16: The eight cycle design principles, with the status of each. Three can be answered today, three depend on the final concept, and two lie outside the project boundary.

	Principle	Status	Evidence or reason
1	No ecotoxic chemicals	**Answered**	Ruled out by constraint; the self-cleaning function must be physical (Section 4.7)
2	Recyclable materials	Depends on V2	**[TODO: state the material of each layer and whether it is separable]**
3	Easy to dismantle, modular	Partly answered	Built to shape and demountable is a stated design rule; the fixing detail is not designed
4	Durability	**Answered**	KPI 7: $\geq 90\,\%$ of the cooling function retained after three years; service life target 20 years
5	Maintainability and reparability	Partly answered	Soft constraint “functional with little maintenance”; no repair concept exists
6	Cascade use of raw materials	**Not addressed**	Nothing in the concept anticipates a second life for its materials
7	Planned return logistics	**Out of boundary**	Requires a business model; Table 3 places this outside the project
8	Efficient use of energy	**Answered**	Hard constraint: no auxiliary energy for the cooling function

Items 6 and 7 are the honest weaknesses of this project from a circular perspective, and they are related. Both need somebody to own the material after its first life, which in turn needs a business model — rental, service or deposit — rather than a sale. The reason is structural and is worth stating because it is not our oversight: in a sales model the manufacturer transfers ownership, so the value created by durable, recoverable design accrues to the recycler, the user and the maintenance trade, and not to the party who paid for it. That split incentive is named as the principal obstacle to circular construction (Ellen MacArthur Foundation 2015), and a façade retrofit has it in an acute form: the owner pays, the tenant gains comfort, the street gains the cooling, and nobody owns the outcome.

4.6 Environmental benefits, limitations and trade-offs

Summer against winter. A high solar reflectance reduces summer heat gain and also reduces useful passive solar gains in winter. The constraint in Table C.2 requires that heating demand must not appreciably rise. **[TODO: Estimate both halves for the Swiss climate and give the net annual balance. Until that number exists this is a known unquantified risk, not a solved trade-off.]**

Cooling against water. Evaporative cooling costs 2.45 MJ per kilogram of water evaporated. For an indicative 100 W/m^2 over the seven-hour load window this is of the order of one litre per square metre and day, so roughly five litres per square metre across the five-day design case — which is within what a 20 mm absorbent layer can hold, and therefore not the binding limit. **[TODO: Recompute with our own assumptions and report a range rather than a point.]** The binding limits are metering, frost, and the fact that during a heat wave it does not rain: the reserve must be charged beforehand.

Local against global. The concept moves heat out of the street canyon. Whether it *reduces* the total energy in the system or merely *redistributes* it is a different question, and the honest answer differs by mechanism: a radiative path sends energy to space and is a genuine removal, while a convective or evaporative path relocates it within the urban boundary layer. **[TODO: State which of these the final concept relies on, and do not claim a reduction where the mechanism only redistributes.]**

Materials. **[TODO: Pigments, binders, any biocide, the separability of the layers at end of life, and the embodied greenhouse gas of the added build-up against the constraint in Table C.2.]**

4.7 Circular does not mean sustainable

Keeping a material in circulation is not by itself an environmental benefit. The standard counterexample is an additive that is harmless in its first application, in a laptop housing, and hazardous in its second after recycling into a children's toy: the substance did not change, the exposure pathway did.

Our project has the same failure mode one step earlier, and it is the reason the life-friendly-chemistry principle is not decorative here. A biocide or a stabiliser that is entirely acceptable *in* a façade coating becomes unacceptable once it leaves that coating, because the path it leaves by is the rainwater this concept deliberately collects and then evaporates at pedestrian level. A substance that would be diluted into a sewer in the incumbent case is, in our case, delivered to the air people breathe. The concept concentrates its own emissions where its beneficiaries are.

Two consequences follow. The self-cleaning function must be structural rather than chemical — which the lichen strategy of Section 3.1 shows is physically possible and the commercial lotus-effect coating cited in Section 2.3 shows is manufacturable. And any substance that does remain in the specification must be checked rather than assumed.

[TODO: Do this check and report it: name the substance, the database, and the result. The REACH registry at the European Chemicals Agency (European Chemicals Agency 2026) is the primary source; PubChem and the EPA DSSTox database are alternatives. One short paragraph naming a substance, a database and a finding is worth more here than a page of general language.]

4.8 Did sustainability considerations change the design?

The template asks this directly, and it is the question a sustainability section most easily fails, because the honest answer is often no. Here it is yes, in three documented instances, each traceable to a decision recorded in Table B.1 in Appendix B.

1. **The reflectance target was lowered from what is technically possible.** Ultra-white coatings reach a solar reflectance of about 0.98 (Li et al. 2021), and an earlier version of this project treated that as the goal. KPI 2 now targets ≥ 0.80, the city's own threshold, because the principle of optimising rather than maximising makes the further reflectance a cost in glare, winter gains and photovoltaic yield rather than a benefit. The performance target was changed by a sustainability argument.
2. **The self-cleaning function was constrained to a physical mechanism.** The obvious solution to soiling is a biocide in the film. The leaching argument of Section 4.7 removed that option before any concept was developed, which is why the lichen strategy entered the model table at all.
3. **Cost and embodied emissions were reclassified.** They were originally an eighth KPI. Recognising that a KPI must follow from a function moved them into the constraints of Table C.2 — where a violation disqualifies a concept rather than merely scoring against it. The change made the requirement stricter, not weaker.

4.9 Limits of this assessment

No life cycle assessment was performed, so no statement in this report should be read as a quantified environmental balance. The embodied emissions of the added build-up are a constraint with a decision rule and not a measured figure. The assessment of Life's Principles above is made against a concept direction rather than a finished design, and **[TODO: must be repeated against Concept V2 once Section 3.3 selects it]**. Finally, two of the eight circular design principles — cascade use and return logistics — are not addressed at all, and the reason is a boundary decision rather than an analysis: they require a business model, and this project does not design one.

5 Discussion & Conclusion

5.1 Contribution of the biomimicry process

[TODO: Be concrete and honest. Which idea came specifically from the biological model, and which would a conventional engineering approach have produced anyway? A well-argued "this part was conventional" is worth more than an overstated claim.]

5.2 Strengths and weaknesses of the final concept

[TODO:]

5.3 Assessment against the challenge and the KPIs

[TODO: Go back through KPIs 1–7 from Table 5 and state for each one whether it was met, missed, or could not be assessed with the available evidence.]

5.4 Innovative character and limits of the evidence

[TODO: State clearly what has been demonstrated (calculation, measurement, literature) and what remains an assumption.]

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Appendix A – AI Use Statement

Relevant AI-supported activities are documented below. Not every prompt is listed; the focus is on how AI contributed to the work and how important outputs were verified. AI output is not used as evidence for factual claims.

Table A.1: AI use statement.

AI tool	**Used for**	**How the output influenced the project**	**How important output was verified**
AskNature (search and strategy pages)	Discovering biological strategies for functions F1–F12, navigated by taxonomy branch rather than by organism	Produced the candidate list in Table 7, including four models that were subsequently de-selected	Every mechanism claim carried into the report is referenced. The pages could not be opened programmatically, so titles and URLs were taken from search results; the open item in Section 3.1 records that each must be confirmed and, where a primary paper exists, replaced by it

AI tool	**Used for**	**How the output influenced the project**	**How important output was verified**
Large language model assistant	Structuring the scoping, distilling the course material into working notes, drafting report text, building the transient wall model of Section 3.2, and generating prompts for figures	Draft text and table structures throughout Sections 1–4; the search-path table; the transient model and its figures	No factual claim entered the report without a named source. Benchmark values were checked against the Fachplanung. Two errors in the AI-built wall model were found and corrected: an air-cavity node made the explicit scheme unstable, and the synthetic weather drive had an inverted diurnal phase, which showed up as a peak surface temperature exactly equal to the air temperature
AI as critical reviewer	**[TODO: Section 3.3 asks for this. Give the reviewer the concept sketch, the Engineering Canvas, the functions and KPIs, and the assumptions]**	**[TODO: Record each comment and the team's decision: accept, reject or investigate]**	**[TODO: A documented *reject* with a reason is worth more here than a list of accepted comments, because it shows judgement]**
[TODO: Further tools used by the team]	**[TODO:]**	**[TODO:]**	**[TODO:]**

On the limits of this statement. Nothing in this report is cited to an AI tool. Where an AI-assisted search surfaced a candidate, the candidate was carried forward only with a source of its own, and the two remaining places where that chain is incomplete are marked as open items rather than left implicit: the AskNature pages listed above, and the 23 % figure for elephant hair, which traces to a university blog rather than to a study (University of Notre Dame 2021).

Appendix B – Decision Log

Every decision below is argued at the place named in the last column. This table exists so that the reasoning can be checked without re-reading the report, and so that what was *not* chosen stays visible: a design process is only legible if the discarded alternatives are on the record.

Table B.1: Decision log. What was chosen, what it displaced, and why.

Decision	**Chosen**	**Rejected alternative**	**Reason, and where it is argued**
Design question	Let the absorbed heat drive its own removal, self-regulating	Four other candidates, from “prevent heat build-up in cities” to “turn heat into a resource”	Too broad, unanchored, factually wrong, or not verifiable in the time available. The chosen question adds the self-regulating clause, which is the actual content of the project. Section 2.1
Reference city	Zurich	A generic European city	An adopted municipal heat strategy, published target values, open geodata and long-term stations make the targets checkable instead of invented. Section 2.2
Reference site	Modellierungsgebiet 03, closed perimeter block	A freely chosen building	The city publishes modelled baseline and climate-optimised results for exactly this area, so the concept is compared against a documented reference. Section 2.2
Design case	Five consecutive hot days	One hot afternoon	Only a multi-day case stresses the water reserve and the reset behaviour of the store; a concept that works on day one and saturates by day three has not solved the problem. Section 2.2
Evening window	18:00–24:00	20:00–24:00	A window must contain the instant at which its own KPI is evaluated, and the two hours before 20:00 are among the hottest of the day at the surface. Section 2.2
System of interest	The façade, with the roof as the release interface	The envelope as a whole, roof included	The roof receives more than twice the noon irradiance, but releases its heat upward to a sky it can see; the façade releases sideways into the canyon at head height, and Section 1 fixes that as where the effect must appear. Section 2.2
Number of systems	One	Two — “reduce heat” and “prevent heat”	The systems view showed these are two functions of one system, coupled by the design question. Section 2.4
Scope of “use the heat”	Let it drive its own removal	Seasonal storage and electricity generation	Conversion competes with photovoltaics for the same surface and any seasonal store lies outside the envelope. Section 2.4

Decision	**Chosen**	**Rejected alternative**	**Reason, and where it is argued**
Selection KPI for the urban effect	PET at 2 m above ground	Urban heat island intensity	A district-scale quantity that a single retrofit cannot move measurably, and it invites the surface-versus-air confusion. Section 2.7
Cost and embodied emissions	Constraints	An eighth KPI	A KPI must follow from a function; these are conditions to satisfy, not measures of performance. The move also brings the list inside the required four to seven. Section 2.7
Albedo target	≥ 0.80, the city's threshold	The technical maximum of ≈ 0.98	Life's Principles ask for optimised rather than maximised; further reflectance costs glare, winter gains and photovoltaic yield. Section 2.8
Life's Principles claimed	Four claimed with the sub-principle named	"Creates opportunities for life" and "Evolve to survive" for the artefact	A damp textured surface invites the algal growth KPI 7 penalises, and a façade layer does not evolve across generations. Naming the conflict is more honest than claiming the principle. Section 2.8
Search strategy	By function, through the taxonomy	By organism	Searching for organisms would not have produced F9, which is the pairing of two taxonomy branches rather than a single one. Section 3.1.1
Termite mound	De-selected	The standard biomimicry reference for passive cooling	The chimney account is contested in the literature, and the building usually cited for it uses fans. Section 3.1
Fog harvesting	De-selected	The Namib beetle, the most-cited water model	Fog in Zurich is a winter phenomenon; a heat wave brings neither rain nor fog. Section 3.1
Green façades	Out of scope	An obvious and proven measure	Already the client's HA 09 and HA 10, and in direct competition with our own water reserve, which is scarcest exactly when planting needs most. Section 3.1
Evidence standard	Primary papers where they exist, the publishing organisation otherwise	AskNature entries treated as evidence in their own right	AskNature and AI-assisted search find candidates. Twelve of twenty model entries still rest on an AskNature page alone; this is recorded as a shortfall rather than presented as met. Section 3.1.1
Concept pair	[TODO: decide]	**[TODO: the two directions not developed]**	**[TODO: state the physical distinction that makes the pair meaningful]**

Decision	Chosen	Rejected alternative	Reason, and where it is argued
Final concept	**[TODO: A, B or a justified hybrid]**	**[TODO:]**	**[TODO: name the weakness in V1 that V2 answers]**

Appendix C – Scoping Detail

Three tables that document the scoping decisions of Section 2.1 to Section 2.4. They are placed here rather than in the body so that Section 2 stays within the page budget while the reasoning behind it remains on record.

Table C.1: Design questions considered, and why four were rejected. Referenced from Section 2.1.

#	Candidate	Assessment
1	How do we prevent heat build-up in urban areas?	*Too broad.* No place, no time, no intervention point — the equivalent of "How might we end hunger?"
2	How could we use the energy or heat build-up positively while lowering temperature?	Right idea, but unanchored: no location, and "positively" does not say for what. Refined into candidate 5.
3	How might we get the heat that Zurich's existing buildings store during the day back out of the city before the next morning, without using energy?	Well formed and highly testable. Rejected only because it discards the team's central idea of *using* the heat. Its night-time focus is retained in the selected question.
4	How can we prevent heat waves causing heat islands?	*Factually wrong.* Heat waves do not cause heat islands. The urban heat island is a permanent consequence of urban surface properties and geometry and exists on ordinary summer days; a heat wave makes it dangerous, it does not create it (Oke 1982).
5	How might we turn the heat that accumulates in a city into a useful resource, instead of reflecting or discarding it?	Good direction, but the reading "store summer heat for winter" is not verifiable within the project, and it has no biological support: organisms store chemical energy across seasons, not sensible heat. Narrowed into the selected question.

Table C.2: Context conditions and constraints in full. Hard constraints disqualify a concept when violated; soft constraints weight the evaluation in Section 3.3. Summarised in Section 2.2.

Type	Class	Constraint
Energy	hard	No auxiliary electrical energy for the cooling function itself. This follows directly from the design question: the heat load must drive the response. Sensors used for validation are excluded.
Water	hard	No potable water. Only rainwater harvested on site or atmospheric moisture.
Existing building	hard	Retrofit onto an existing envelope. Load-bearing structure, floor plan and window openings are not redesigned.
Townscape	hard	The result must be acceptable in the Zurich townscape. Buildings may be listed; heritage protection, building law and neighbour rights can restrict visible changes (Stadt Zürich 2020a, p. 128).
Glare	hard	No specular glare towards streets, neighbouring flats or traffic. The city names glare as the main objection to bright envelopes (Stadt Zürich 2020a, p. 128), which favours diffuse over mirror-like surfaces.
Climate	hard	Must survive the Swiss annual cycle: frost to about $-10\,°\mathrm{C}$, freeze–thaw cycles, driving rain, UV, and periods of several dry weeks in summer.
Winter	hard	Must not appreciably increase heating demand. A high solar reflectance also reduces useful winter solar gains; this trade-off must be quantified, not ignored.
Installation	soft	Must be installable by ordinary trades on an occupied building, ideally without full scaffolding. **[TODO: state an installation-time target per m^2]**
Maintenance	soft	Must remain functional with little maintenance. Soiling and the resulting cleaning cost are named by the city as a specific weakness of high-albedo envelopes (Stadt Zürich 2020a, p. 128). Service life target ≥ 20 years.
Solar energy	soft	Should not unnecessarily compromise roof- or façade-integrated photovoltaics; the city notes that bright envelopes reduce PV yield (Stadt Zürich 2020a, p. 128). State which envelope areas the concept claims.
Structural	soft	Limited additional dead load and build-up thickness on an existing envelope.
Embodied emissions	soft	The greenhouse gas embodied in the added build-up must be repaid by the avoided cooling demand within the service life. Measured in kg CO_2-eq $1/\mathrm{m}^2$. **[TODO: source a benchmark]**
Economic	soft	The city expects higher cost for high-albedo envelopes (Stadt Zürich 2020a, p. 128). Target: cost per square metre in the range of a normal refurbishment. **[TODO: find a benchmark figure and cite it]**

Table C.3: What the comparison with each parallel system contributed to the scoping. Referenced from Figure 1 and Section 2.4.

Parallel system	**What the comparison adds**
Stormwater and sewer system	Rainwater is currently drained away as fast as possible. Seen from our system it is a cooling reserve that is being discarded — and during a heat wave it is the only water available under the "no potable water" constraint. This adds the functions *capture*, *retain* and *meter* water (F6–F8) and makes storage capacity a design variable rather than an afterthought.
Street and pavement network	The street is treated by the same client with the same logic (HA 06, Table 2), and it achieves a slightly larger daytime effect than the envelope. The comparison sets a benchmark we must beat, and it shows that the envelope's advantage has to come from the night window, where the street performs no better.
Power grid and building energy system	The obvious way to "use" heat is to convert it. The comparison shows why that is a different project: conversion competes with photovoltaics for the same surface, and any seasonal store lies outside the envelope. It is therefore placed out of the boundary, while the milder reading of "use" — letting the heat drive its own removal — stays in.
Residents, transport and public transit	Defines where the effect has to be delivered: at pedestrian level in the street canyon and in the flats on the upper floors, not on the roof membrane. This is why PET at 2 m above ground becomes a KPI, and it fixes the evening window 18:00–24:00, when people are outdoors and then sleeping.